Math 105L In Class 2 Practice Solutions

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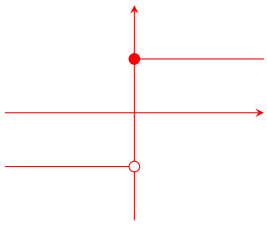
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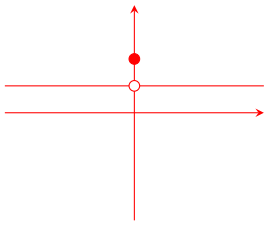
Limits and Continuity

1. Describe the three ways in which a function can fail to be continuous. Draw an example in each case.

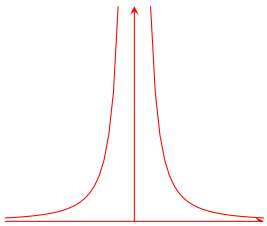
   - We might not be able to know what to expect:



   - We might not get what we expect:



   - What we expect might not be a number:



2. We said that $f(x)$ is *continuous* at $x = a$ if $\lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x) = f(a)$.

   We also said that $f(x)$ is continuous at $x = a$ if we both *know what to expect at* $x = a$ and *get what we expect at* x=a.

   Of the two equalities in the first definition, which says we know what to expect, and which says we get it? Explain.

   $\lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x)$ says we know what to expect. The two sides approach the same number. That is, $\lim_{x\to a} f(x)$ exists.

   If, *in addition to that*, $\lim_{x\to a} f(x) = f(a)$, we also get what we expect, since the actual value of the function is what we expect it to be.

3. Consider the function $f(x) = \frac{x^2-1}{x+1}$.

   (a) Explain why $f(x)$ is not continuous at $x = -1$.

   $f(-1)$ gives $\frac{0}{0}$, so the function is not defined there. Therefore, regardless of the limits from each side, we cannot get what we expect.

   (b) Consider

$$g(x) = \begin{cases} f(x) & x \neq -1 \\ k & x = -1 \end{cases}$$

   Is there a value of k that makes $g(x)$ continuous? If so, find it. If not, explain why.

   We want $\lim_{x\to -1^+} f(x) = \lim_{x\to -1^-} f(x) = f(-1) = k$:

$$\begin{aligned} \lim_{x\to -1^+} f(x) &= \lim_{x\to -1^+} \frac{x^2-1}{x+1} \\ &= \lim_{x\to -1^+} \frac{(x+1)(x-1)}{x+1} \\ &= \lim_{x\to -1^+} x - 1 = -2 \end{aligned}$$

   (the limit from the negative side is the same).

   So $k = -2$ makes this continuous.

4. Consider the function

$$f(x) = \begin{cases} -1 & x > 0 \\ k & x = 0 \\ 1 & x < 0 \end{cases}$$

   Is there a value of k that makes $f(x)$ continuous? If so, find it. If not, explain why.

   We want $\lim_{x\to 0^+} f(x) = \lim_{x\to 0^-} f(x) = f(0) = k$:

   $\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} -1 = -1$

   $\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} 1 = 1$

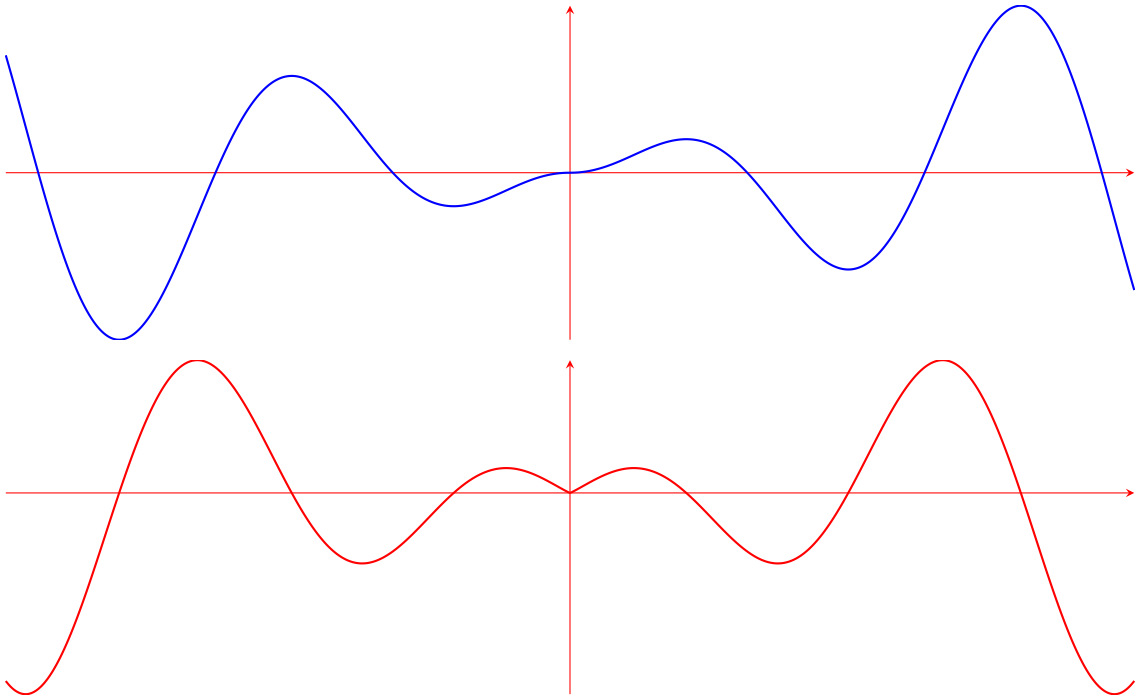
   Since these are not equal, there is no value of k that would make $f(x)$ continuous.

5. Briefly explain why it makes sense that $\lim_{x\to a} kf(x) = k \lim_{x\to a} f(x)$.

   If we know what value we expect $f(x)$ to take at $x = a$ (i.e $\lim_{x\to a} f(x)$ exists), then we expect the function $kf(x)$ to take a value that is k times larger there, since $kf(x)$ just multiplies all y-values by k.

Derivative Function

1. Sketch the graph of the derivative of the following function on the axes below.



2. If $f(x)$ is positive from $x = 0$ to $x = 3$, and negative from $x = 3$ to $x = 6$, what is the sign of its derivative from $x = 3$ to $x = 6$? Explain.

   There's no way to know. Being positive or negative has nothing to do with the derivative.

3. Suppose that $F(p)$ is a function giving the number of thousands of fish in a county with p ponds in it. Explain what the following quantities mean:

   (a) $F(20) = 100$.

   There are 100,000 fish in a county with 20 ponds.

   (b) $F'(20) = 6$.

   If a county has 20 ponds, then adding another one increases the number of fish in the county by 6,000.

4. Let $A(t)$ be the depth below sea level of submarine t hours after its launch, in feet.

   (a) Explain why the statement $A(5) = -1000$ makes no sense. Include units in your answer.

   It would mean that the submarine is -1000 ft below sea level, or 1000 ft up in the air.

   (b) Suppose that $A'(5) = -500$. Discuss the motion of the submarine at this time. Include units.

   The submarine is moving up *toward the surface* at 500 ft/hr.

5. If $f(x) = x^2 + 2x$, compute $f'(x)$ *from the definition of the derivative.*

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{((x+h)^2 + 2(x+h)) - (x^2+2x)}{h} \\ &= \lim_{h\to 0} \frac{x^2 + 2xh + h^2 + 2x + 2h - x^2 - 2x}{h} \\ &= \lim_{h\to 0} \frac{2xh + h^2 + 2h}{h} \\ &= \lim_{h\to 0} \frac{h(2x + h + 2)}{h} \\ &= \lim_{h\to 0} 2x + h + 2 = 2x + 2 \end{aligned}$$

Second Derivative

1. If $f''(x) > 0$, then which of the following is true? Explain why each is true, false, or unknowable.

   (a) The function $f(x)$ is positive.

   (b) The tangent lines to $f(x)$ are increasing.

   (c) The slopes of the tangent lines to $f(x)$ are increasing.

   (a) and (b) are both unknowable. The function may or may not be positive. The tangent lines could be increasing ($f'(x) > 0$) or decreasing ($f'(x) < 0$), but we have no idea from this.

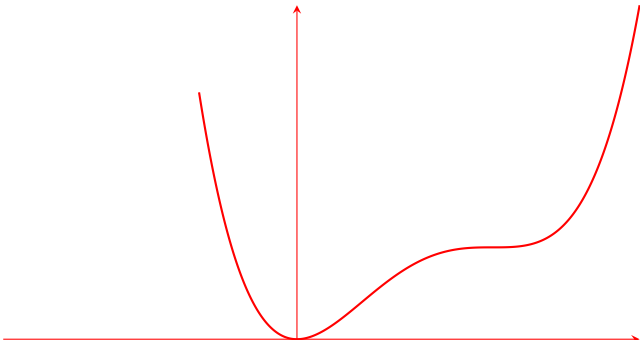
   (c) is true. $f''(x) > 0$ means exactly that: as x increases, the slopes of the tangent lines increase too, since $f''(x)$ is the rate of change of the slopes.

2. Sketch graphs of smooth continuous functions with domain all real numbers matching each one of the following descriptions, or explain why it's impossible.

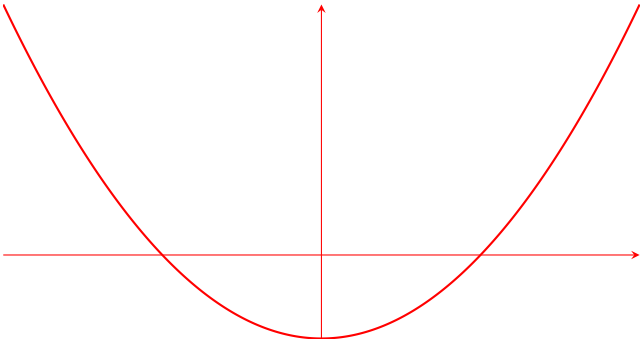
   (a) $f(x) > 0$, $f'(x) < 0$, $f''(x) < 0$, and $\lim_{x\to\infty} = 5$.

   Impossible. If the function is decreasing and concave down, it gets steeper as x increases, meaning it can't have a HA at $y = 5$.

   (b) $f(x)$ has two inflection points, $f'(x) < 0$ for $x < 0$, $f'(x) \geq 0$ for $x > 0$.



   (c) $f(x)$ has no inflection points, $f'(x) = 0$ at exactly one point, $f(x)$ has two zeros.



3. True or False? Explain your answers.

   (a) If a function is concave up, then $f''(x) > 0$.

   False. $f(x) = x^4$ is concave up, but has $f''(x) = 12x^2$, which is 0 at $x = 0$.

(b) If a function is concave down, then $f''(x) \leq 0$.

True. If at any point, $f''(x) > 0$, $f(x)$ is concave up there, so if $f(x)$ is concave down, $f''(x)$ must be non-positive.

(c) When a function turns around, it has an inflection point.

False. Turning points and inflection points are not the same thing.

(d) It is possible for a function to be decreasing, but have increasing slopes.

True. This just means it's decreasing and concave up. So for example, e^{-x} for all real numbers, or x^2 for $x < 0$.

4. Fill in the blanks: If a function is increasing, then its derivative or slope is positive. If the slopes of a function are decreasing, then its second derivative is negative.

5. The position of an object above the ground at time t is given by a function $s(t)$, where $s'(t) < 0$ and $s''(t) > 0$. Is the object rising or falling? Is it speeding up or slowing down? Explain.

The object is falling, since its height is decreasing ($s'(t) < 0$).

The object is slowing down: its velocity is negative, but increasing (since $s''(t) > 0$, and $s''(t)$ is the rate of change of $s'(t)$, the velocity). So its velocity is becoming less negative, which is closer to zero. Therefore the object is slowing down.

Product and Quotient Rules

1. Differentiate $f(x) = x^2(3x^3 + 1)$.

   Use the product rule. Let $u(x) = x^2$, $v(x) = 3x^3 + 1$.

   $$u'(x) = 2x, \quad v'(x) = 9x^2$$

   $$f'(x) = u'v + uv' = 2x(3x^3 + 1) + x^2(9x^2)$$

2. Differentiate $g(x) = (x^3 - 2x)(x + 5)$.

   Product rule: $u(x) = x^3 - 2x$, $v(x) = x + 5$.

   $$u'(x) = 3x^2 - 2, \quad v'(x) = 1$$

   $$g'(x) = u'v + uv' = (3x^2 - 2)(x + 5) + (x^3 - 2x)(1)$$

3. Differentiate $h(x) = \sqrt{x}\,(2x^2 - 1)$. (Assume $x > 0$.)

   Write $\sqrt{x} = x^{1/2}$. Let $u(x) = x^{1/2}$, $v(x) = 2x^2 - 1$.

   $$u'(x) = \tfrac{1}{2}x^{-1/2}, \quad v'(x) = 4x$$

   $$h'(x) = u'v + uv' = \tfrac{1}{2}x^{-1/2}(2x^2 - 1) + x^{1/2}(4x)$$

4. Differentiate $p(x) = (x - 1)^2(x^2 + 2)$.

   Let $u(x) = (x - 1)^2$, $v(x) = x^2 + 2$.

   $$u'(x) = 2(x - 1), \quad v'(x) = 2x$$

   $$p'(x) = u'v + uv' = 2(x - 1)(x^2 + 2) + (x - 1)^2(2x)$$

5. Differentiate $q(x) = e^x(x^2 - 3x + 1)$.

   Let $u(x) = e^x$, $v(x) = x^2 - 3x + 1$.

   $$u'(x) = e^x, \quad v'(x) = 2x - 3$$

   $$q'(x) = u'v + uv' = e^x(x^2 - 3x + 1) + e^x(2x - 3)$$

6. Differentiate $r(x) = \dfrac{x^2 + 1}{x}$.

   Quotient rule: $u(x) = x^2 + 1$, $v(x) = x$.

   $$u'(x) = 2x, \quad v'(x) = 1$$

   $$r'(x) = \frac{u'v - uv'}{v^2} = \frac{2x(x) - (x^2 + 1)(1)}{x^2}$$

7. Differentiate $s(x) = \dfrac{3x-1}{x^2+1}$.

Let $u(x) = 3x-1$, $v(x) = x^2+1$.

$$u'(x) = 3, \quad v'(x) = 2x$$

$$s'(x) = \frac{u'v - uv'}{v^2} = \frac{3(x^2+1) - (3x-1)(2x)}{(x^2+1)^2}$$

8. Differentiate $t(x) = \dfrac{x^3+2}{x-1}$.

Quotient rule: $u(x) = x^3+2$, $v(x) = x-1$.

$$u'(x) = 3x^2, \quad v'(x) = 1$$

$$t'(x) = \frac{u'v - uv'}{v^2} = \frac{3x^2(x-1) - (x^3+2)(1)}{(x-1)^2}$$

9. Differentiate $u(x) = \dfrac{\sqrt{x}}{x+2}$.

Write $\sqrt{x} = x^{1/2}$. Let $p(x) = x^{1/2}$, $q(x) = x+2$.

$$p'(x) = \tfrac{1}{2}x^{-1/2}, \quad q'(x) = 1$$

$$u'(x) = \frac{p'q - pq'}{q^2} = \frac{\frac{1}{2}x^{-1/2}(x+2) - x^{1/2}(1)}{(x+2)^2}$$

10. Differentiate $v(x) = (x^2+1)(2x^3-x)$.

Let $a(x) = x^2+1$, $b(x) = 2x^3-x$.

$$a'(x) = 2x, \quad b'(x) = 6x^2-1$$

$$v'(x) = a'b + ab' = 2x(2x^3-x) + (x^2+1)(6x^2-1)$$

11. Find the tangent line to $f(x) = x^2(3x+1)$ at $x = 1$.

Use the product rule. Let $u(x) = x^2$, $v(x) = 3x+1$.

$$u'(x) = 2x, \quad v'(x) = 3$$

$$f'(x) = u'v + uv' = 2x(3x+1) + x^2(3)$$

Evaluate at $x = 1$:

$$f(1) = 1^2(3\cdot 1+1) = 4, \quad f'(1) = 11$$

Point-slope form:

$$y - f(1) = f'(1)(x-1) \quad \text{so} \quad y - 4 = 11(x-1)$$

Final tangent line:

$$y = 4 + 11(x-1)$$

12. Find the tangent line to $g(x) = (x^3 - 2x)(x + 2)$ at $x = 0$.

Product rule. Let $u(x) = x^3 - 2x$, $v(x) = x + 2$.

$$u'(x) = 3x^2 - 2, \quad v'(x) = 1$$

$$g'(x) = u'v + uv' = (3x^2 - 2)(x + 2) + (x^3 - 2x)(1)$$

At $x = 0$:

$$g(0) = 0, \quad g'(0) = -4$$

Point-slope form:

$$y - g(0) = g'(0)(x - 0) \quad \text{so} \quad y - 0 = -4(x - 0)$$

Final tangent line:

$$y = -4x$$

13. Find the tangent line to $h(x) = \sqrt{x}(2x^2 - 1)$ at $x = 1$.

Write $\sqrt{x} = x^{1/2}$. Let $u(x) = x^{1/2}$, $v(x) = 2x^2 - 1$.

$$u'(x) = \frac{1}{2}x^{-1/2}, \quad v'(x) = 4x$$

$$h'(x) = u'v + uv' = \frac{1}{2}x^{-1/2}(2x^2 - 1) + x^{1/2}(4x)$$

At $x = 1$:

$$h(1) = 1, \quad h'(1) = \frac{9}{2}$$

Point-slope form:

$$y - h(1) = h'(1)(x - 1) \quad \text{so} \quad y - 1 = \frac{9}{2}(x - 1)$$

Final tangent line:

$$y = 1 + \frac{9}{2}(x - 1)$$

14. Find the tangent line to $p(x) = (x - 1)^2(x^2 + 1)$ at $x = 2$.

Let $u(x) = (x - 1)^2$, $v(x) = x^2 + 1$.

$$u'(x) = 2(x - 1), \quad v'(x) = 2x$$

$$p'(x) = u'v + uv' = 2(x - 1)(x^2 + 1) + (x - 1)^2(2x)$$

At $x = 2$:

$$p(2) = 5, \quad p'(2) = 14$$

Point-slope form:

$$y - p(2) = p'(2)(x - 2) \quad \text{so} \quad y - 5 = 14(x - 2)$$

Final tangent line:

$$y = 5 + 14(x - 2)$$

15. Find the tangent line to $q(x) = e^x(x^2 - 2x)$ at $x = 0$.

Let $u(x) = e^x$, $v(x) = x^2 - 2x$.

$$u'(x) = e^x, \quad v'(x) = 2x - 2$$

$$q'(x) = u'v + uv' = e^x(x^2 - 2x) + e^x(2x - 2)$$

At $x = 0$:

$$q(0) = 0, \quad q'(0) = -2$$

Point-slope form:

$$y - q(0) = q'(0)(x - 0) \quad \text{so} \quad y - 0 = -2(x - 0)$$

Final tangent line:

$$y = -2x$$

16. Find the tangent line to $r(x) = \frac{x^2+1}{x}$ at $x = 1$.

Quotient rule. Let $u(x) = x^2 + 1$, $v(x) = x$.

$$u'(x) = 2x, \quad v'(x) = 1$$

$$r'(x) = \frac{u'v - uv'}{v^2} = \frac{2x \cdot x - (x^2 + 1)}{x^2}$$

At $x = 1$:

$$r(1) = 2, \quad r'(1) = 0$$

Point-slope form:

$$y - r(1) = r'(1)(x - 1) \quad \text{so} \quad y - 2 = 0 \cdot (x - 1)$$

Final tangent line:

$$y = 2$$

17. Find the tangent line to $s(x) = \frac{3x-1}{x^2+1}$ at $x = 1$.

Let $u(x) = 3x - 1$, $v(x) = x^2 + 1$.

$$u'(x) = 3, \quad v'(x) = 2x$$

$$s'(x) = \frac{u'v - uv'}{v^2} = \frac{3(x^2 + 1) - (3x - 1)(2x)}{(x^2 + 1)^2}$$

At $x = 1$:

$$s(1) = 1, \quad s'(1) = \frac{1}{2}$$

Point-slope form:

$$y - s(1) = s'(1)(x - 1) \quad \text{so} \quad y - 1 = \frac{1}{2}(x - 1)$$

Final tangent line:

$$y = 1 + \frac{1}{2}(x - 1)$$

18. Find the tangent line to $t(x) = \frac{x^3+1}{x-1}$ at $x = 2$.

Let $u(x) = x^3 + 1$, $v(x) = x - 1$.

$$u'(x) = 3x^2, \quad v'(x) = 1$$

$$t'(x) = \frac{u'v - uv'}{v^2} = \frac{3x^2(x - 1) - (x^3 + 1)}{(x - 1)^2}$$

At $x = 2$:

$$t(2) = 9, \quad t'(2) = 3$$

Point-slope form:

$$y - t(2) = t'(2)(x - 2) \quad \text{so} \quad y - 9 = 3(x - 2)$$

Final tangent line:

$$y = 9 + 3(x - 2)$$

19. Find the tangent line to $u(x) = \frac{\sqrt{x}}{x+1}$ at $x = 4$.

Write $\sqrt{x} = x^{1/2}$. Let $p(x) = x^{1/2}$, $q(x) = x + 1$.

$$p'(x) = \frac{1}{2}x^{-1/2}, \quad q'(x) = 1$$

$$u'(x) = \frac{p'q - pq'}{q^2} = \frac{\frac{1}{2}x^{-1/2}(x+1) - x^{1/2}}{(x+1)^2}$$

At $x = 4$:

$$u(4) = \frac{2}{5}, \quad u'(4) = -\frac{3}{100}$$

Point-slope form:

$$y - u(4) = u'(4)(x - 4) \quad \text{so} \quad y - \frac{2}{5} = -\frac{3}{100}(x - 4)$$

Final tangent line:

$$y = \frac{2}{5} - \frac{3}{100}(x - 4)$$

20. Find the tangent line to $v(x) = (x^2 + 1)(2x^2 - x)$ at $x = -1$.

Let $a(x) = x^2 + 1$, $b(x) = 2x^2 - x$.

$$a'(x) = 2x, \quad b'(x) = 4x - 1$$

$$v'(x) = a'b + ab' = 2x(2x^2 - x) + (x^2 + 1)(4x - 1)$$

At $x = -1$:

$$v(-1) = 6, \quad v'(-1) = -16$$

Point-slope form:

$$y - v(-1) = v'(-1)(x + 1) \quad \text{so} \quad y - 6 = -16(x + 1)$$

Final tangent line:

$$y = 6 - 16(x + 1)$$

Compound Interest

1. Which of the following is good approximation of $Ae^{0.04t}$? Explain your answer.

   (a) $A\left(1+\frac{1}{0.04}\right)^{1,000,000}$

   (b) $A\left(1+\frac{0.04}{1,000,000}\right)^{1,000,000}$

   (c) $A\left(1+\frac{1,000,0000}{0.04}\right)^{1,000,000}$

   (d) $A\left(1+\frac{1}{0.04}\right)^{0.04}$

   (b) is the right answer. We know that $Ae^{0.04t}$ is the amount of money you have if you invest $\$A$ in a bank account with 4% interest, compounded continuously. (b) gives the same, but compounded $1,000,000$ times a year, which is a very close approximation of continuous compounding.

2. "The more often you compound, the more money you make, so it's worth insisting on an initial investment of \$1,000 being compounded every second rather than every minute."

   Critique this statement.

   Not really. While it's true that compounding every second technically results in a slightly higher number, the difference is many decimal points down the line and would not materially affect the amount of money. For example, after 10 years at 5% interest, every minute would get

   $$1000\left(1+\frac{0.05}{365\times 24\times 60}\right)^{365\times 24\times 60\times 10}\approx 1648.72$$

   , and every second

   $$1000\left(1+\frac{0.05}{365\times 24\times 60\times 60}\right)^{365\times 24\times 60\times 60\times 10}\approx 1648.72$$

   So we get the same amount of money.

3. How much money would you have if you invested \$1000 at 50% annual interest, compounded twice a year? Don't use a formula here: show computations from scratch.

   25% interest, added twice. After once, we have $\$1,000 * 1.25 = 1\$1,250$. After twice, $\$1,250 * 1.25 = \$1,562.50$.

4. Write an equation whose solution gives the amount of money you'd need to invest to have \$100,000 in 20 years, assuming you put it in a bank account with continuously compounded 6% interest.

   If A is the amount of money you need to put in, then $100,000 = Ae^{0.06\times 20}$.

5. Suppose that you have the opportunity either get \$10,000 in ten years, or \$5,000 now, given that you can put the money in a bank account with 8% interest, compounded continuously. Which would you prefer? Explain.

   If you take the second option, you'd end up with $5000e^{0.08\times 10}\approx \$11,127$. Definitely better.

Differentiability

1. True or false? Justify your answer.

   (a) If a function is continuous at $x = a$, then it is also differentiable there.

   False. For example, $|x|$ is continuous at $x = 0$, but not differentiable there.

   (b) Suppose that a function has a vertical tangent line at a point $x = a$. Then since it has a tangent line there, it's differentiable there.

   False. A vertical tangent line would mean the derivative would be $\pm\infty$. The definition of differentiability requires a finite derivative.

   (c) If a function is differentiable at $x = a$, it is continuous there.

   True. We have a theorem that says that.

2. A student was asked to check if the function

$$f(x) = \begin{cases} x^2 + 1 & x \leq 0 \\ x^3 & x > 0 \end{cases}$$

   is differentiable at $x = 0$. What is wrong with the following answer?

   "We differentiate away from $x = 0$ to get

$$f'(x) = \begin{cases} 2x & x < 0 \\ 3x^2 & x > 0 \end{cases}$$

   Then

$$\lim_{x \to 0^+} f'(x) = \lim_{x \to 0^+} 3x^2 = 0$$

   and

$$\lim_{x \to 0^-} f'(x) = \lim_{x \to 0^-} 2x = 0$$

   The two limits are the same, so $f'(x)$ is continuous at $x = 0$, making $f(x)$ differentiable at $x = 0$."

   The student forgot to check for continuity of $f(x)$. If we do, we get

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} x^3 = 0$$

   and

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} x^2 + 1 = 1$$

   So the function is not continuous and therefore not differentiable.

3. Use the limit definition of the derivative (no shortcut!) to show that the function

$$f(x) = \begin{cases} 2x + 7 & x > 1 \\ x^2 + 8 & x \leq 1 \end{cases}$$

   is differentiable.

   We need to check at $x = 1$. $f(1) = 1^2 + 8 = 9$.

$$\begin{aligned}\lim_{h\to 0^+}\frac{f(1+h)-f(1)}{h} &= \lim_{h\to 0+}\frac{(2(1+h)+7)-9}{h}\\ &= \lim_{h\to 0^+}\frac{2+2h+7-9}{h}\\ &= \lim_{h\to 0^+}\frac{2h}{h}\\ &= \lim_{h\to 0^+} 2 = 2\end{aligned}$$

$$\begin{aligned}\lim_{h\to 0^-}\frac{f(1+h)-f(1)}{h} &= \lim_{h\to 0-}\frac{((1+h)^2+8)-9}{h}\\ &= \lim_{h\to 0^-}\frac{1+2h+h^2+8-9}{h}\\ &= \lim_{h\to 0^-}\frac{2h+h^2}{h}\\ &= \lim_{h\to 0^-}\frac{h(2+h)}{h}\\ &= \lim_{h\to 0^-} 2+h = 2\end{aligned}$$

4. Is there a value a so that the function

$$f(x) = \begin{cases} x^3+ax & x>1\\ ax^2+1 & x\le 1\end{cases}$$

is differentiable at $x=1$? If so, find such a value.

Continuity: $\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} x^3+ax = 1+a$, and $\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} ax^2+1 = a+1$.

So the function is continuous for all a.

$f'(x) = \begin{cases} 3x^2+a & x>1\\ 2ax & x<1\end{cases}$.

$\lim_{x\to 1^+} f'(x) = \lim_{x\to 1^+} 3x^2+a = 3+a$ and $\lim_{x\to 1^-} f'(x) = \lim_{x\to 1^-} 2ax = 2a$.

We need $3+a=2a$, or $a=3$. That makes the function differentiable.

5. A function can fail to be differentiable in three ways. Name them and give an example of each.

Corner: $|x|$ at $x=0$.

Vertical tangent line: $\sqrt[3]{x}$ at $x=0$.

Discontinuity: $f(x) = \begin{cases} 1 & x>0\\ -1 & x\le 0\end{cases}$.

Exponential Functions

1. Which of the following functions is exponential?

   (a) $f(x) = 3x + 2$
   (b) $f(x) = x^3$
   (c) $f(x) = 2^x$
   (d) $f(x) = e^{x-1}$
   (e) $f(x) = 7^{x^2}$
   (f) $f(x) = |x| + 7$
   (g) $f(x) = x^x$
   (h) $f(x) = e^x + x$

   An exponential function is one in which the variable appears only in the exponent, not as a base or added outside the exponential term. Hence, the exponential functions here are:

   $$f(x) = 2^x \quad \text{and} \quad f(x) = e^{x-1} = e^{-1} \cdot e^x.$$

2. Suppose that $f(x) = P \cdot a^x$, with $a > 0$ and $a \neq 1$. Which of the following are true regardless of the values of P and a (subject to $a > 0$, $a \neq 1$)? For those that are not always true, give values of P and a that make them true.

   (a) $f(x)$ is increasing.
   (b) $f(x)$ is decreasing.
   (c) $f(x)$ is concave up.
   (d) $f(x)$ is concave down.

   None are always true.

   (a) True for (eg) $a > 1$ and $P > 0$, so $f(x) = 2^x$, eg. Also for $P < 0$ and $a < 1$, so $f(x) = -(0.5)^x$, eg
   (b) True for (eg) $a < 1$ and $P > 0$, so $f(x) = 0.5^x$, eg. Also for $P < 0$ and $a > 1$, so $f(x) = -(2^x)$, eg
   (c) True for any a with $P > 0$, so eg $f(x) = 2^x$ or $f(x) = 0.5^x$.
   (d) True for any a with $P < 0$, so eg $f(x) = -(2^x)$ or $f(x) = -(0.5^x)$.

3. Write the function $22e^{0.2t}$ in the form $P_0 a^t$.

   $22e^{0.2t} = 22(e^{0.2})^t \approx 22 \times (1.221)^t$

4. Suppose that a radioactive substance has a half life of 4 days. If we start with 20 moles, find a formula of the form $P_0 a^t$ for the number of moles left at time t days, and use it to find how much is left after 6 days. Show all your work – no shortcut formulas.

   $P_0 = 20$ moles.

   After 4 days, we have 10 moles left, so $10 = 20a^4$, giving $a = \left(\frac{1}{2}\right)^{0.25} \approx 0.841$.

   So $N(t) = 20(0.841)^t$, giving $N(6) \approx 20(0.841)^6 \approx 7.1$ moles.

5. Suppose that $a > 0$. What coordinate do the graphs of all functions of the form $f(x) = a^x$ pass through?

   $f(0) = a^0 = 1$, regardless of the value of a, so all such functions pass through $(0, 1)$.

Logs

1. Find the quadrupling time of a bank account with 6% continuously compounded interest.

   We use the continuous compounding formula: $A = Pe^{rt}$

   We want to find the time t such that $A = 4P$.

$$\begin{aligned} 4P &= Pe^{0.06t} \\ 4 &= e^{0.06t} \\ \ln 4 &= 0.06t \\ t &= \frac{\ln 4}{0.06} \\ t &\approx \frac{1.386294}{0.06} \\ t &\approx 23.10 \text{ years.} \end{aligned}$$

   The quadrupling time is approximately 23.10 years.

2. (a) Find the inverse of $f(x) = 3e^{2x} - 1$.

   Set $y = 3e^{2x} - 1$. Then

$$y + 1 = 3e^{2x} \quad \Rightarrow \quad e^{2x} = \frac{y+1}{3}$$

$$2x = \ln\left(\frac{y+1}{3}\right) \quad \Rightarrow \quad x = \tfrac{1}{2}\ln\left(\frac{y+1}{3}\right).$$

   Hence the inverse function is

$$f^{-1}(x) = \tfrac{1}{2}\ln\left(\frac{x+1}{3}\right).$$

   (b) Find the inverse of $g(x) = \ln(5x - 2)$.

   Let $y = \ln(5x - 2)$. Then

$$e^y = 5x - 2 \quad \Rightarrow \quad x = \frac{e^y + 2}{5}.$$

   Therefore

$$g^{-1}(x) = \frac{e^x + 2}{5},$$

   (c) Find the inverse of $h(x) = 2^{x+3} + 4$.

   Take $y = 2^{x+3} + 4$.

   Then $y - 4 = 2^{x+3}$.

   So $y - 4 = \ln(2^{x+3}) = (x + 3)\ln 2$

   $x + 3 = \frac{\ln(y-4)}{\ln 2}$

   $x = \frac{\ln(y-4)}{\ln 2} - 3$

   So $h^{-1}(x) = \frac{\ln(x-4)}{\ln 2} - 3$ (valid for $x > 4$).

(d) Find the inverse of $p(x) = \frac{1}{4}\ln(x) + 7$.

Let $y = \frac{1}{4}\ln(x) + 7$. Then

$$y - 7 = \tfrac{1}{4}\ln(x) \quad \Rightarrow \quad \ln(x) = 4(y-7)$$

$$x = e^{4(y-7)}.$$

So the inverse is

$$p^{-1}(x) = e^{4(x-7)},$$

(e) Find the inverse of $q(x) = \ln\big(2 - e^{-x}\big)$.

Start with $y = \ln\big(2 - e^{-x}\big)$. Then

$$e^y = 2 - e^{-x} \quad \Rightarrow \quad e^{-x} = 2 - e^y.$$

$$-x = \ln(2 - e^y) \quad \Rightarrow \quad x = -\ln(2 - e^y).$$

Therefore

$$q^{-1}(x) = -\ln\big(2 - e^x\big),$$

3. (a) Solve $5^x = 12$.

Take natural logs of both sides:

$$\ln(5^x) = \ln 12 \quad \Rightarrow \quad x \ln 5 = \ln 12.$$

Hence

$$x = \frac{\ln 12}{\ln 5}.$$

(b) Solve $10^{2x+1} = 7 \cdot 10^{x-2}$.

Divide both sides by 10^{x-2}:

$$10^{(2x+1)-(x-2)} = 7 \quad \Rightarrow \quad 10^{x+3} = 7.$$

Taking log (base 10) gives

$$x + 3 = \log 7 \quad \Rightarrow \quad x = \log 7 - 3.$$

(c) Solve $9^{7-2x} + 6 = e$.

Start with

$$9^{7-2x} + 6 = e.$$

Then

$$9^{7-2x} = e - 6.$$

But $e - 6 < 0$, whereas $9^{7-2x} > 0$ for every real x. Therefore there is no real solution to the equation.

(d) Solve $\log(2x+3) + \log(x-1) = 1$. (Give exact answer and check if your answers are possible!)

Domain requires $2x + 3 > 0$ and $x - 1 > 0$, so $x > 1$. Combine the logs:

$$\log\big((2x+3)(x-1)\big) = 1 \quad \Rightarrow \quad (2x+3)(x-1) = 10.$$

Expand and solve the quadratic:

$$2x^2 + x - 3 = 10 \quad \Rightarrow \quad 2x^2 + x - 13 = 0.$$

Use the quadratic formula:

$$x = \frac{-1 \pm \sqrt{1 + 104}}{4} = \frac{-1 \pm \sqrt{105}}{4}.$$

Only the root greater than 1 is admissible. Since $\frac{-1+\sqrt{105}}{4} > 1$ and $\frac{-1-\sqrt{105}}{4} < 1$, the solution is

$$x = \frac{-1 + \sqrt{105}}{4}.$$

(e) Solve $\log(4x) - \log(x+1) = \frac{1}{2}$. (Give exact answer and check if your answers are possible.)

Domain: $4x > 0$ and $x + 1 > 0$, so $x > 0$. Combine logs:

$$\log\left(\frac{4x}{x+1}\right) = \tfrac{1}{2} \quad \Rightarrow \quad \frac{4x}{x+1} = 10^{1/2} = \sqrt{10}.$$

Solve the resulting linear equation:

$$4x = \sqrt{10}\,(x+1) \quad \Rightarrow \quad 4x = \sqrt{10}\,x + \sqrt{10}$$

$$(4 - \sqrt{10})x = \sqrt{10} \quad \Rightarrow \quad x = \frac{\sqrt{10}}{4 - \sqrt{10}}.$$

This x is positive (so it satisfies the domain restriction), and is the exact solution.

4. Let $x = \log A$ and $y = \log B$. Rewrite each of the following in terms of x and y:

(a) $\log \sqrt{\dfrac{A}{B}}$

$$\log \sqrt{\frac{A}{B}} = \log \left(\frac{A}{B}\right)^{1/2} = \tfrac{1}{2}\big(\log A - \log B\big) = \tfrac{1}{2}(x - y).$$

(b) $\log\left(\dfrac{A^3\sqrt{B}}{10}\right)$

$$\log\left(\frac{A^3\sqrt{B}}{10}\right) = \log A^3 + \log B^{1/2} - \log 10 = 3\log A + \tfrac{1}{2}\log B - 1 = 3x + \tfrac{1}{2}y - 1.$$

(c) $\log\left(\dfrac{\sqrt[3]{A}\,B^{-2}}{A^{-1}}\right)$

First simplify the argument:

$$\frac{\sqrt[3]{A}\,B^{-2}}{A^{-1}} = A^{1/3}B^{-2}A^1 = A^{4/3}B^{-2}.$$

Hence

$$\log\big(A^{4/3}B^{-2}\big) = \tfrac{4}{3}\log A - 2\log B = \tfrac{4}{3}x - 2y.$$

(d) $\dfrac{\log\left(A^2B\right)}{\log\left(\sqrt{AB}\,\right)}$

Evaluate numerator and denominator:

$$\log\left(A^2B\right) = 2\log A + \log B = 2x + y,$$

$$\log\left(\sqrt{AB}\,\right) = \tfrac{1}{2}\log(AB) = \tfrac{1}{2}(\log A + \log B) = \tfrac{1}{2}(x+y).$$

Therefore

$$\frac{\log\left(A^2B\right)}{\log\left(\sqrt{AB}\,\right)} = \frac{2x+y}{\frac{1}{2}(x+y)} = \frac{2(2x+y)}{x+y}.$$

5. Write 9^x in the form 8^{kx}.

$9^x = 8^{kx}$, so $\ln 9^x = \ln 8^{kx}$.

$x \ln 9 = kx \ln 8$, or $k = \frac{\ln 9}{\ln 8} \approx 1.0566$.

So $9^x \approx 8^{1.0566x}$.

Chain Rule

1. Find the derivatives of the following functions:

(a) $y = (2x^3 - x)^5$.

Let $f(x) = x^5$, $g(x) = 2x^3 - x$. Then $f'(x) = 5x^4$, $g'(x) = 6x^2 - 1$. Thus,

$$\frac{dy}{dx} = f'(g(x))g'(x) = 5(2x^3 - x)^4(6x^2 - 1).$$

(b) $y = e^{3x^2+5}$.

Let $f(x) = e^x$ and $g(x) = 3x^2 + 5$. Then $f'(x) = e^x$, $g'(x) = 6x$. So,

$$\frac{dy}{dx} = f'(g(x))g'(x) = e^{3x^2+5}(6x).$$

(c) $y = ((x^2 + 1)^3 + 2)^4$.

(Triple Chain Rule) Let $f(x) = x^4$, $g(x) = x^3+2$, $h(x) = x^2+1$. Then $f'(x) = 4x^3$, $g'(x) = 3x^2$, $h'(x) = 2x$. Therefore,

$$\frac{dy}{dx} = f'(g(h(x)))g'(h(x))h'(x) = 4((x^2 + 1)^3 + 2)^3(3(x^2 + 1)^2)(2x).$$

Simplify:

$$\frac{dy}{dx} = 24x((x^2 + 1)^3 + 2)^3(x^2 + 1)^2.$$

(d) $y = (e^{x^3+3x^2+2x})^2$.

(Triple Chain Rule) Let $f(x) = x^2$, $g(x) = e^x$, $h(x) = x^3 + 3x^2 + 2x$. Then $f'(x) = 2x$, $g'(x) = e^x$, $h'(x) = 3x^2 + 6x + 2$. So,

$$\frac{dy}{dx} = f'(g(h(x)))g'(h(x))h'(x) = 2(e^{x^3+3x^2+2x})(e^{x^3+3x^2+2x})(3x^2 + 6x + 2).$$

Simplify:

$$\frac{dy}{dx} = 2e^{2(x^3+3x^2+2x)}(3x^2 + 6x + 2).$$

(e) $y = (x^4 + 2x^2 + 3)^{1/3}$.

Let $f(x) = x^{1/3}$, $g(x) = x^4 + 2x^2 + 3$. Then $f'(x) = \frac{1}{3}x^{-2/3}$, $g'(x) = 4x^3 + 4x$. Hence,

$$\frac{dy}{dx} = f'(g(x))g'(x) = \frac{1}{3}(x^4 + 2x^2 + 3)^{-2/3}(4x^3 + 4x).$$

(f) $y = e^{(4x^3+1)^{1/2}}$.

(Triple Chain Rule) Let $f(x) = e^x$, $g(x) = x^{1/2}$, $h(x) = 4x^3 + 1$. Then $f'(x) = e^x$, $g'(x) = \frac{1}{2}x^{-1/2}$, $h'(x) = 12x^2$. So,

$$\frac{dy}{dx} = f'(g(h(x)))g'(h(x))h'(x) = e^{(4x^3+1)^{1/2}}\frac{1}{2}(4x^3 + 1)^{-1/2}(12x^2).$$

Simplify:

$$\frac{dy}{dx} = \frac{6x^2e^{(4x^3+1)^{1/2}}}{(4x^3 + 1)^{1/2}}.$$

(g) $y = (5x^4 - 7x^2 + 2)^3$.

Let $f(x) = x^3$, $g(x) = 5x^4 - 7x^2 + 2$. Then $f'(x) = 3x^2$, $g'(x) = 20x^3 - 14x$. Thus,

$$\frac{dy}{dx} = f'(g(x))g'(x) = 3(5x^4 - 7x^2 + 2)^2(20x^3 - 14x).$$

(h) $y = ((x^2 + 1)^4 + 3x)^5$.

(Triple Chain Rule) Let $f(x) = x^5$, $g(x) = x^4 + 3x$, $h(x) = x^2 + 1$. Then $f'(x) = 5x^4$, $g'(x) = 4x^3 + 3$, $h'(x) = 2x$. Therefore,

$$\frac{dy}{dx} = f'(g(h(x)))g'(h(x))h'(x) = 5((x^2 + 1)^4 + 3x)^4(4(x^2 + 1)^3(2x) + 3(2x)).$$

(i) $y = e^{(x^2+1)^3}$.

(Triple Chain Rule) Let $f(x) = e^x$, $g(x) = x^3$, $h(x) = x^2 + 1$. Then $f'(x) = e^x$, $g'(x) = 3x^2$, $h'(x) = 2x$. Hence,

$$\frac{dy}{dx} = f'(g(h(x)))g'(h(x))h'(x) = e^{(x^2+1)^3}(3(x^2 + 1)^2)(2x).$$

Simplify:

$$\frac{dy}{dx} = 6x(x^2 + 1)^2 e^{(x^2+1)^3}.$$

(j) $y = (3x^2 + (x + 1)^3)^4$.

(Triple Chain Rule) Let $f(x) = x^4$, $g(x) = 3x^2 + x^3 + 3x^2 + 3x + 1$, $h(x) = x + 1$. Then $f'(x) = 4x^3$, $g'(x) = 6x + 3(x + 1)^2$, $h'(x) = 1$. Thus,

$$\frac{dy}{dx} = f'(g(h(x)))g'(h(x))h'(x) = 4(3x^2 + (x + 1)^3)^3(6x + 3(x + 1)^2).$$

2. Supose that $f(x)$ is a differentiable function. Differentiate the following. Most involve the triple chain rule.

(a) $f(x)^9$

Let the outer function be $F(x) = x^9$ and the inner be $f(x)$. Then

$$\frac{d}{dx} f(x)^9 = F'(f(x)) \cdot f'(x) = 9f(x)^8 f'(x).$$

(b) $f\big((x^4 + 2)^5\big)$

(Triple Chain Rule)

Let $f(g(h(x)))$ where $g(x) = x^5$, $h(x) = x^4 + 2$. Then

$$\frac{d}{dx} f\big((x^4 + 2)^5\big) = f'\big((x^4 + 2)^5\big) \cdot 5(x^4 + 2)^4 \cdot 4x^3$$

$$= 20x^3(x^4 + 2)^4 f'\big((x^4 + 2)^5\big).$$

(c) $e^{f(x)^2+1}$

(Triple Chain Rule)

Let $F(x) = e^x$, $g(x) = x^2 + 1$. So $F'(x) = e^x$, and $g'(x) = 2x$, giving:

$$\frac{d}{dx}e^{f(x)^2+1} = e^{f(x)^2+1} \cdot (2f(x)f'(x)).$$

Hence

$$\frac{dy}{dx} = 2f(x)f'(x)e^{f(x)^2+1}.$$

(d) $2^{f(x)^3}$

(Triple Chain Rule)

Let $F(x) = 2^x$ and $g(x) = f(x)^3$. Innermost is $f(x)$ Then $F'(x) = 2^x \ln 2$. Thus

$$\frac{d}{dx}2^{f(x)^3} = 2^{f(x)^3} \ln(2) \cdot 3f(x)^2 f'(x).$$

(e) $\big(f(x)^4 + x\big)^8$

(Triple Chain Rule)

Outer $F(x) = x^8$, $g(x) = f(x)^4 + x$. Innermost is $f(x)$ Then

$$\frac{d}{dx}\big(f(x)^4 + x\big)^8 = 8\big(f(x)^4 + x\big)^7 \cdot (4f(x)^3 f'(x) + 1).$$

(f) $f\big((x^2 + 1)^7\big)$

(Triple Chain Rule)

Outermost is $f(x)$. Let $g(x) = x^7$, $h(x) = x^2 + 1$,

$$\frac{dy}{dx} = f'\big((x^2 + 1)^7\big) \cdot 7(x^2 + 1)^6 \cdot 2x = 14x(x^2 + 1)^6 f'\big((x^2 + 1)^7\big).$$

(g) $(10^{f(x)} + 1)^5$

(Triple Chain Rule)

Let $F(x) = x^5$, $G(x) = 10^x + 1$. Innermost is $f(x)$ Then $F'(x) = 5x^4$, $G'(x) = 10^x \ln 10$. Hence

$$\frac{dy}{dx} = 5(10^{f(x)} + 1)^4 (10^{f(x)} \ln 10) f'(x).$$

(h) $f\big(e^{x^3+2x}\big)$

(Triple Chain Rule)

Outermost is $f(x)$. Let $g(x) = e^x$, $h(x) = x^3 + 2x$. Then

$$\frac{d}{dx}f(e^{x^3+2x}) = f'(e^{x^3+2x}) \cdot e^{x^3+2x} \cdot (3x^2 + 2).$$

(i) $(f(x)^5 + 2)^9$

(Triple Chain Rule)

Outer $F(x) = x^9$, middle $G(x) = x^5 + 2$, inner $f(x)$. Then

$$\frac{dy}{dx} = 9(f(x)^5 + 2)^8 \cdot 5f(x)^4 f'(x) = 45f(x)^4 f'(x)(f(x)^5 + 2)^8.$$

(j) $f\left((x^4+3x^2+1)^3\right)$

(Triple Chain Rule)

Outermost is $f(x)$. With $g(x)=x^3$, $h(x)=x^4+3x^2+1$,

$$\frac{dy}{dx} = f'\left((x^4+3x^2+1)^3\right)\cdot 3(x^4+3x^2+1)^2(4x^3+6x).$$

3. Let $f(x)$ and $g(x)$ be differentiable. Find a formula for the derivative of $\frac{f(x)}{[g(x)]^2}$. Then ***use it*** to compute the derivative of $\frac{e^x}{x^4}$.

$$\begin{aligned}\frac{d}{dx}\frac{f(x)}{[g(x)]^2} &= \frac{d}{dx}f(x)[g(x)]^{-2}\\ &= f'(x)[g(x)]^{-2} + f(x)\frac{d}{dx}[g(x)]^{-2}\end{aligned}$$

$\frac{d}{dx}[g(x)]^{-2}$: Inner function $g(x)$, outer $f(x)=x^{-2}$.

So $f'(x)=-2x^{-3}$, giving $\frac{d}{dx}[g(x)]^{-2} = -2[g(x)]^{-3}g'(x)$.

Plugging back in:

$$\begin{aligned}\frac{d}{dx}\frac{f(x)}{[g(x)]^2} &= f'(x)[g(x)]^{-2} + f(x)\frac{d}{dx}[g(x)]^{-2}\\ &= f'(x)[g(x)]^{-2} + f(x)\times -2[g(x)]^{-3}g'(x)\\ &= \frac{f'(x)}{g(x)^2} - \frac{2f(x)g'(x)}{g(x)^3}\\ &= \frac{f'(x)g(x) - 2f(x)g'(x)}{g(x)^3}\end{aligned}$$

Let $f(x)=e^x$ and $g(x)=x^2$, then $\frac{f(x)}{g(x)^2} = \frac{e^x}{x^4}$. Also, $f'(x)=e^x$ and $g'(x)=2x$.

We get:

$$\begin{aligned}\frac{d}{dx}\frac{e^x}{x^4} &= \frac{e^x\cdot x^2 - 2e^x\cdot 2x}{x^6}\\ &= \frac{xe^x(x-4)}{x^6}\\ &= \frac{e^x(x-4)}{x^5}\end{aligned}$$

4. Suppose that the depth of a lake in feet at distance d feet from its center is give by a function $f(d)$. Suppose also that the density of fish (in fish/ft^3) in an area of the lake with depth D feet is given by a function $p(D)$.

   (a) Explain what the function $F(d)=p(f(d))$ means.

   It gives the density of fish (in fish/ft^3) at a distance d feet from the center of the lake.

(b) Explain why the function $f(p(d))$ doesn't make sense.

The output of $p(d)$ is density, in fish/ft^3. The input to $f(d)$ is distance in feet. Those don't match up.

(c) What are the units of $f'(d)$, $p'(D)$, and $F'(d)$?

$f'(d)$: Feet of depth per foot of distance from the center.

$p'(D)$: Fish/ft^3 per foot of depth.

$F'(d)$: Fish/ft^3 per foot of distance from the center.

(d) Suppose that:

- at a distance of $1,000$ feet from the center of the lake, the depth of the lake is 20 feet, and is getting shallower at a rate of 0.1 ft of depth for each foot further from the center.
- When the depth is 20 ft, for each ft deeper, the density of fish increases by 100 fish/ft^3.

Compute the rate at which the density of fish increases or decreases for every foot you go further out from the center when you are $1,000$ feet from the center. Give units.

This is $F'(1,000)$.

First bullet point says that $f(1000) = 20$ft, and that $f'(1000) = -0.1$ feet of depth/foot of distance from the center.

This is negative, since as distance from the center increases, the depth gets shallower.

Second one says that $p'(20) = 100$ fish/ft^3/ft of depth.

$$\begin{aligned} F'(1,000) &= p'(f(1,000))f'(1,000) \\ &= p'(20)f'(1,000) \\ &= 100 \times -0.1 \\ &= -10 \text{ fish/ft}^3\text{/ft of distance from the center} \end{aligned}$$

So for every foot you move further out, the fish density *decreases* by 10 fish/ft^3.

Differentiating Exponentials

Note: some of the chain rule problems above cover derivatives of exponentials.

1. Fill in the blanks: Let $a > 0$. Then the derivative of $f(x) = a^x$ is directly proportional to $f(x)$ with constant of proportionality $\ln a$.

2. Suppose that $e^{g(x)} = x$. By differentiating both side of this equation, show that $g'(x) = \frac{1}{x}$.

   Chain rule. Inner function is $g(x)$, outer is $f(x) = e^x$, whose derivative is $f'(x) = e^x$.

   So $f'(g(x)) = e^{g(x)}$.

   $\frac{d}{dx}e^{g(x)} = e^{g(x)}g'(x)$ and $\frac{d}{dx}x = 1$.

   So $e^{g(x)}g'(x) = 1$. Since $e^{g(x)} = x$, we get $xg'(x) = 1$, or $g'(x) = \frac{1}{x}$.

3. Find the slope of the tangent line to $f(x) = 2^x + e^x + e^2 + 2^e + x^2 + x^e$ at $x = 1$.

   $f'(x) = \ln 2 \cdot 2^x + e^x + 2x + ex^{e-1}$.

   Slope is $f'(1) = 2\ln 2 + e + 2 + e = 2\ln 2 + 2e + 2$.

4. Find a function of the form $f(x) = a^x$, so that $f'(x) = -f(x)$.

   $\frac{d}{dx}a^x = \ln a \cdot a^x$.

   This must equal $-a^x$, so we get $\ln a \cdot a^x = -a^x$, or $\ln a = -1$, or $a = e^{-1}$.

5. What is the exact value of $\lim_{h\to 0} \dfrac{7^h - 1}{h}$? Explain.

   $\lim_{h\to 0} \dfrac{7^h - 1}{h} = \ln 7$.

   We know that $\frac{d}{dx}7^x = (\ln 7)7^x$. Computing the same, we get:

$$\begin{aligned}\frac{d}{dx}7^x &= \lim_{h\to 0}\frac{7^{x+h} - 7^x}{h}\\ &= \lim_{h\to 0}\frac{7^x 7^h - 7^x}{h}\\ &= \lim_{h\to 0}\frac{7^x(7^h - 1)}{h}\\ &= 7^x \lim_{h\to 0}\frac{7^h - 1}{h}\end{aligned}$$

   Therefore $7^x \lim_{h\to 0}\frac{7^h-1}{h} = (\ln 7)7^x$. Divide by 7^x to get the result.